
CS 203: Discrete Structures

Course Code: CS 203

Course Name: Discrete Structures

L-T-P-C: 3-0-0-3

Students intended for 2nd year B. Tech. CSE

Elective or Compulsory: Compulsory

Prerequisites: None

Approval: 3rd Senate

Course Contents

- Logics and Proofs: Propositional Logic, Applications of Propositional Logic, Propositional Equivalences, Predicates and Quantifiers, Nested Quantifiers, Rules of Inference, Introduction to Proofs, Proof Methods and Strategies.
- Sets, Functions, Sequences, Sum, and Matrices: Sets, Set operators, Functions, Sequences and Summations, Cardinality of Sets, Matrices
- Counting and Discrete Probability: Basics of Counting, Pigeonhole Principle, Permutations and Combinations, Introduction to Discrete Probability, Probability Theory, Bayes Theorem, Expected Value and Variance
- Relations: Relations and Their Properties, n-ary Relations and Their Applications, Representing Relations, Closures of Relations, Equivalence Relations, Partial Orderings
- Graphs: Graphs and Graph Models, Representing Graphs and Graph Isomorphism, Connectivity, Euler and Hamilton Paths, Shortest-Path Problems, Planar Graphs, Graph Coloring
- Trees: Introduction to Trees, Application of Trees, Tree Traversal, Spanning Trees, Minimum Spanning Trees
- Boolean Algebra and Modeling Computation: Boolean Algebra, Representing Boolean Functions, Logic Gates, Minimization of Circuits, Language and Grammars, Finite-State Machines, Language Recognition, Turing Machines

Readings (including but not restricted to the following):

1. Kenneth Rosen [KR], **Discrete mathematics and its applications**, 6th Edition, McGraw-Hill Science/Engineering/Math, 2006.

Reference Books:

1. C. Liu, D. Mohapatra[CM], **Elements of Discrete Mathematics**, Tata McGraw-Hill, 2008.
2. T.Koshy [TK], **Discrete mathematics with applications**, Academic Press, 2003
3. J. Hein [JH], **Discrete structures, logic and computability**, Jones & Bartlett Publishers, 2009.